The Delta Confluence

How SahayakMap Guided NDRF Command Through the Crisis

A technical narrative illustrating SahayakMap's operational intelligence pipeline in action during a simulated MBB delta flood event — 17 May 2026.

Introduction: The Edge of Chaos

The rain over the Odisha delta wasn't falling; it was driving horizontally — a relentless grey wall that had turned the Mahanadi, Brahmani, and Baitarani river basins into a single, contiguous inland sea. Inside the Regional Command Center, Commander Rajesh Sharma of the National Disaster Response Force (NDRF) stared at the main projection wall.

In previous monsoons, this room was an echo chamber of panic. WhatsApp groups would flash with unverified screenshots, local officials would jam phone lines with contradictory reports, and deployment maps were littered with post-it notes representing rescue assets sent blind into the storm.

But today, 17 May 2026, was different. Behind Rajesh's console, humming silently on a remote VPS instance (**sahayak-map-app.service**), sat **SahayakMap** — an operational flood-intelligence dashboard running a synchronized network of automated pipelines. Rajesh wasn't drowning in data anymore; he was commanding a tactical data link.

Chapter 1: The Social Net and Hydrologic Truth

At 10:14 AM, the first anomaly arrived not from an official gauge, but from the digital streets. Deep within the system's ingestion framework, **social_agent.py** was executing its automated 2-hour cycle. Its X API v2 keyword matrix was scanning the public stream with station-geocoded queries, filtering noise using -is:retweet constraints to locate ground-truth signals.

A tweet from an unverified handle in Jajpur district entered the pipeline:

```
"Brahmani river rising crazy fast near Jenapur bridge. NH16 traffic moving very
slow. If water rises 1m more road will be cut off! #Jajpur #OdishaFlood"
```

The pipeline immediately initiated the **Fake Detection and Triangulation Protocol**:

1. Spatial Anchoring: Keywords *Jenapur*, *NH16*, and *Jajpur* routed the record to the JEN station query bucket.

2. Meteorological Cross-Reference: The parser fetched live IMD Nowcast data via /api/vps/feeds. Jajpur district was at **wms_color 3 (Alert: Orange)**.

3. Sensor Cross-Reference: The system queried Supabase gauge_readings for JEN. WSE was reading 24.2m — climbing toward Danger Level (23.0m already breached).

The LLM extraction returned **severity=4, reliability_score=4** — passing the verification gate (severity >= 3, reliability >= 3, IMD wms_color >= 3). The signal was marked **verified_crisis=True** and classified as *road_closure / P1_critical*.

On Rajesh's dashboard, the amber Jenapur marker pulsed. The alert panel fired Rule R6:

```
ROUTE VIABILITY ALERT: NH-16 (Coastal Corridor) near Jenapur Bridge — severe
riverine pressure. Structural inundation predicted within 3 hours.
```

Rajesh adjusted his headset. *"Command to Jajpur Highway Patrol. Restrict heavy vehicle movement on NH-16 near the Baitarani-Brahmani interface. Divert logistics to the interior bypass."* A critical logistics artery was saved before the water even touched the asphalt.

Chapter 2: The Silent Crisis and the 2G Fail-Safe

By 1:30 PM, the crisis escalated into a Silent Emergency. The telemetry feed for **Purusottampur** (PUR) went flatline. Cell towers were collapsing under extreme wind loads.

Under active flood states (WSE >= Warning), **Rule R8** had already tightened its stale threshold from 48 hours to an 8-hour window. PUR hadn't pinged in exactly 8 hours. The intelligence engine escalated the station to **Priority 1 (Comm-Blackout)**, turning the Leaflet marker grey.

Within that blackout zone, Team Delta-7 (B-05) was navigating roaring currents near Mundali Barrage when their outboard motor died. Their field terminal dropped to a single bar of **2G EDGE**.

As the browser's Network Information API detected effectiveType === '2g', the dashboard executed its **UI Stripping Transformation**: Leaflet map killed, heavy assets purged, the application pivoted to the /api/pulse endpoint — a pipe-delimited compact text packet of 149 bytes:

```
NAR,26.5,DANGER,3,B01|JEN,24.5,WARNING,2,B02|PUR,0.0,STALE,9,--|...
```

Fields: CODE, WSE, STATUS, LEVEL, NEAREST_BOAT. The crew saw a high-contrast text-only tactical grid consuming near-zero bandwidth. They tapped the high-contrast SOS button.

The SOS transmitted as a Telegram message through the bot's hybrid polling architecture. The crew's coordinates, identity, and timestamp reached **telegram_bot.py** on the VPS.

Chapter 3: Automated Triage and the Dispatch

Rajesh's Telegram console chimed. The backend's **triage_logic.py** had already intercepted the SOS coordinates (20.4650, 85.9450) and executed its analytical loop:

1. Haversine Proximity Scan: Filtered boat_assets for status IN ('Safe','Active','En Route') AND last_ping within 2 hours.

2. Candidate Selection: B-01 (Delta-1) at 3.2km, B-10 (Tango-8) at 6.8km.

3. Route Risk Check: Cross-referenced path against road_risks table. SH-42 flagged as High Risk. Alternate route recommended.

4. Safe Zone: Cuttack 6th Bn ODRAF Base recommended from safe_zones table.

Rajesh's screen showed:

```
SOS DISPATCH: Team Delta-7 (B-05) stalled at Mundali Barrage. Position: 20.4650,
85.9450 | District: Cuttack Nearest Assets: 1. B-01 (Delta-1) - 3.2km [Safe] 2.
B-10 (Tango-8) - 6.8km [Safe] ROUTE CAUTION: SH-42 High Risk. Use elevated
embankment road. Safe Zone: Cuttack 6th Bn ODRAF Base [Approve Dispatch] [Override
Plan]
```

Rajesh tapped **Approve**. The Telegram bot saved state['last_sos']['status'] = 'dispatched'. On the EOC dashboard, the dashed red route line from B-01 to the SOS point solidified — signalling an authorized rescue operation underway. *Note: route solidification requires the /api/sos/active endpoint to expose the status field — this is a known near-term improvement.*

Chapter 4: The Intel Chat Under Pressure

At 2:45 PM, a senior EOC coordinator unfamiliar with the dashboard tapped the floating **SahayakMap Intel** chat button (bottom-right). She typed:

```
"Is there any storm approaching the delta?"
```

The /api/chat endpoint routed the intent to **radar**. It downloaded the Paradip Doppler MAX-Z GIF in base64, passed it to **Groq llama-4-scout** (vision model) alongside the compressed live gauge context (6 stations, stale excluded), and returned:

```
"Storm cells detected northwest of the delta with light rain (blue/green
reflectivity). Stations NAR, JEN, ANA, AKH reporting normal WSE. No immediate
threat to delta operations."
```

She then asked: *"Which WhatsApp message is critical?"* The bot replied: *"Cuttack — Knee-deep water near Naraj — severity 4 is critical."*

And: *"Are the tweets about Jenapur bridge real or fake?" — "VERIFIED: NH-16 at risk, corroborated by gauge and IMD. LIKELY FAKE: single unverified source claiming 100 stranded with no gauge support."*

Epilogue: The Defensible Audit Trail

By 6:00 PM, the storm's peak had broken. Team B-05 had been rescued by B-01 and safely disembarked at Cuttack 6th Bn ODRAF Base. NH-16 had remained passable for relief convoys.

A state audit official arrived requesting justification for asset re-allocation and road closures. Rajesh clicked **Export EOC SITREP**.

pdf_export.py queried the append-only alerts table and social_signals ledger, compiling every timestamped event into a timestamp-locked PDF. The report detailed:

```
10:14 AM: Twitter signal (severity=4, reliability=4) — Jenapur bridge NH-16 risk. Corroborated: JEN WSE 24.2m + IMD Orange Alert. Rule R6 fired. 1:30 PM: Rule R8 triggered — PUR silent 8h during active flood. Priority 1 escalation. 1:32 PM: Telegram SOS from B-05 at 20.4650, 85.9450. Triage: B-01 dispatched (3.2km), SH-42 avoided, ODRAF safe zone confirmed.
```

The official closed his folder and nodded. *"Fully transparent and defensible."*